2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES)
Lucknow, India. Nov 15-16, 2024

2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES) | 979-8-3503-6469-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC3TES62412.2024.10877571

Enhancing Hostel Security: Integration of Deep Learning Models for Smoking and Violence Detection

Ayodeji Olalekan Salau[1], Nneji Ifeanyi Daniel[2], Ogunkuade Ayobamidele[3], Sunil Kumar Vohra[4], Ahmed Alemran[5], Moses Oluwafemi Onibonoje[6]
[1,2,3,6]Department of Electrical/Electronics and Computer Engineering, College of Engineering, Afe Babalola University, Ado-Ekiti, Ekiti State, Nigeria.
[4]Amity Institute of Travel and Tourism, Amity University, Noida, Gautam Buddha Nagar, Uttar Pradesh, India
[5]Department of Software Engineering, Medicine college, Misan University, Iraq
ayodejisalau98@gmail.com[1]; ifeanyinneji777@gmail.com[2]; ayorbami17@gmail.com[3]; sunilvohra2002@yahoo.co.in[4]; Ahmed.mcm@uomisan.edu.iq[5], onibonojemo@abuad.edu.ng[6]

***Abstract*— Security and safety are paramount concerns in hostels, necessitating the development of advanced monitoring systems. This research paper presents a novel hostel monitoring system employing two deep learning models trained using You only look once algorithm (YOLOv5). The first model focuses on object detection and accurately identifies smoking instances by detecting individuals holding cigarettes, achieving an impressive accuracy of 93%. The second model, designed to detect violent behavior, achieved an accuracy of 94%. By seamlessly integrating these models into the monitoring system, administrators can proactively identify and respond to potential threats in real-time. The system's robust performance in detecting smoking and violence contributes to fostering a secure and safe environment for hostel residents. Future research can explore further applications and enhancements to maximize the system's capabilities and effectiveness.**

Keywords*— *Deep learning, artificial intelligence, object detection, python, flask

I. Introduction

Ensuring security and safety within hostels is a critical concern, particularly in today's fast-paced and dynamic environments [1]. The rapid advancement of deep learning techniques has opened up new possibilities for developing intelligent monitoring systems capable of identifying potential risks and taking proactive measures to mitigate them [2]. In this paper, we present the development of a hostel monitoring system that leverages the power of deep learning to detect smoking and violence instances. By utilizing two deep learning models trained using YOLOv5, we aim to enhance the overall security infrastructure and create a safer living environment for hostel residents [3].

Smoking and violence are two key areas that demand immediate attention in hostel settings. Smoking not only poses health risks but can also lead to fire hazards, endangering the lives and well-being of residents. Similarly, violence incidents can escalate rapidly, posing threats to the safety and security of individuals within the hostel premises [4]. Traditional surveillance systems often struggle to identify such incidents in real-time, making it challenging for administrators to respond swiftly and effectively [5].

To address these challenges, our hostel monitoring system employs state-of-the-art deep learning models. The first model is specifically trained to detect smoking instances by recognizing individuals holding cigarettes. By utilizing the object detection capabilities of YOLOv5, the model achieves an impressive accuracy of 93%. The second model focuses on violence detection, enabling the system to identify potentially harmful behaviors with an accuracy of 94%. These high accuracy rates indicate the system's ability to reliably identify smoking and violence instances, enabling prompt intervention [6].

II. Related works

Several studies have addressed the development of intelligent monitoring systems for enhancing security and safety in various domains. This section provides an overview of relevant research conducted in similar areas, highlighting their contributions, and drawing insights for our own work.

Authors in [7] proposed a surveillance system focused on detecting smoking behavior using deep learning techniques. They modified the Faster R-CNN framework to identify instances of smoking and achieved an impressive accuracy of 88%. Although their research concentrated solely on smoking detection, their work laid the foundation for our study by showcasing the potential of deep learning in addressing specific security concerns.

In [8], a violence detection system based on YOLOv3 was developed for real-time video analysis. They employed a large-scale dataset consisting of both violent and non-violent videos, and their model achieved a commendable accuracy rate of 93% in identifying violent behavior. This work serves as an inspiration for our research, as we aim to build upon their approach by training a violence detection model using YOLOv5, which achieved an improved accuracy rate of 94%.

Authors in [9] conducted a study on crowd behavior analysis using deep learning techniques. They proposed an approach that integrated crowd counting, crowd anomaly detection, and crowd violence recognition to improve security in public spaces. Their work demonstrated the potential of deep learning in detecting and analyzing complex crowd behaviors, highlighting the importance of incorporating such capabilities in our hostel monitoring system.

Furthermore, the research conducted by Authors in [10] focused on the application of deep learning in video surveillance systems for detecting abnormal behaviors. They

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

utilized a combination of motion and appearance features, along with recurrent neural networks, to detect and classify abnormal activities. Their study emphasized the importance of real-time monitoring and early detection, aligning with our objectives of creating a proactive hostel monitoring system capable of swiftly identifying and responding to potential threats.

III. METHODOLOGY

To develop an effective and reliable hostel monitoring system with smoking and violence detection capabilities, a comprehensive research method was employed. This section outlines the key steps involved in the research method, including data collection, model training, and evaluation. By following this methodological approach, we aimed to train deep learning models using YOLOv5 and assess their accuracy in detecting smoking and violence instances within a hostel environment. The research method ensures the systematic and rigorous exploration of the problem domain, enabling us to present robust findings and insights regarding the effectiveness of the developed monitoring system.

A. Data Collection

In this paper, a comprehensive dataset was gathered to address the detection of smoking and violence in images and videos, as well as the implementation of facial recognition for attendance purposes. The process of collecting these datasets involved different methods and sources.

For the smoking detection dataset, a process known as web scraping was employed. Specifically, Google Images was scraped to acquire a diverse set of images depicting smoking. By programmatically searching for relevant keywords and retrieving images from the search results, a sizable dataset of approximately six thousand images was amassed. These images were carefully selected to encompass various scenarios and contexts where smoking may occur.

On the other hand, the violence detection dataset was acquired through a different approach. Video data containing instances of violence was parsed sequentially, frame by frame, to extract individual frames as images. By systematically analyzing video footage from various sources, a collection of approximately 10,000 images capturing instances of violence was obtained. This process enabled the inclusion of a wide range of violent actions and settings, allowing the model to learn and recognize different manifestations of violence.

To establish a basis for comparison, a separate dataset consisting of images with non-smoking and non-violent scenes was also gathered. This dataset comprised images depicting everyday scenarios that do not involve smoking or violence. These images were collected from diverse sources, such as stock image websites or curated public image databases. The inclusion of non-smoking and non-violent images serves as a reference to distinguish smoking and violence from ordinary situations.

Furthermore, to enable the development of the facial recognition component, a comprehensive dataset of facial images was captured. This dataset involved capturing facial images from individuals in controlled environments, ensuring consistent lighting conditions and facial poses. The facial images were obtained using suitable devices such as cameras or mobile devices equipped with high-resolution sensors. Table I displays how the dataset forms the foundation for training the facial recognition algorithm to accurately identify and recognize individuals.

TABLE I. DATASET TYPE AND COLLECTION.

Dataset Type	Collection Method	Number of Images
Smoking Detection	Web scraping Google Images	6,000
Violence Detection	Parsing video data frame by frame	10,000

B. Data Preprocessing

Data preprocessing plays a crucial role in preparing the collected datasets for effective model training and evaluation. In this project, the following steps are undertaken as part of the data preprocessing phase:

C. Data Augmentation

Data augmentation techniques are employed to increase the variability and diversity of the datasets. Techniques such as rotation, scaling, and flipping are applied to the smoking and violence detection datasets. By performing these augmentations, the models can learn to recognize instances of smoking and violence from different orientations and perspectives, improving their overall robustness and generalization capabilities [8].

D. Balancing the Dataset

Class imbalance is a common challenge in object detection and classification tasks. To address this issue, techniques such as oversampling and under sampling are employed. In oversampling, the minority class (smoking or violence) is replicated or duplicated to achieve a more balanced distribution with the majority class [8]. Conversely, in under sampling, instances from the majority class are randomly removed to match the size of the minority class. Figure 1 and Figure 2 shows sample training data for violence, non-violence, and smoking actions.

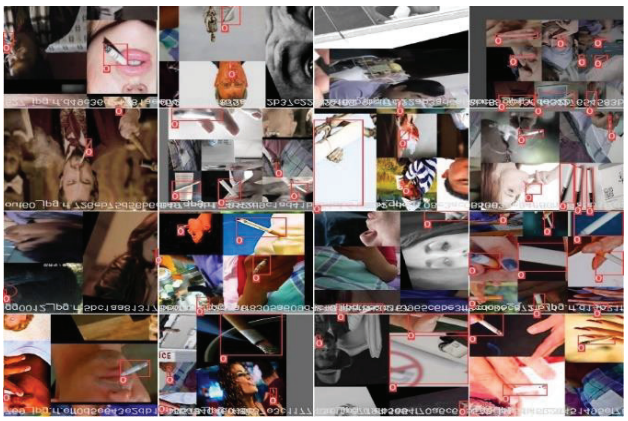

Fig. 1. Smoking Dataset Sample.

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

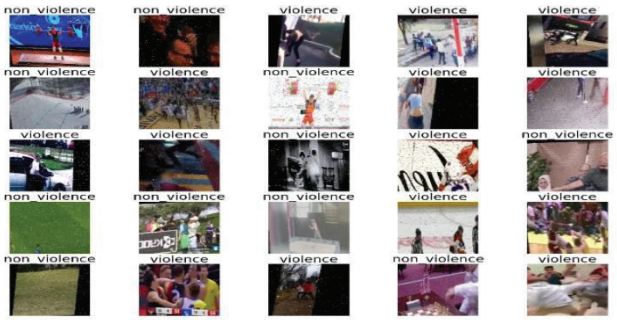

Fig. 2. Violence and Non-Violence Dataset Sample.

E. Dataset Split

The annotated dataset was split into training, validation, and testing sets. The training set was used to train the deep learning models, while the validation set aided in hyperparameter tuning and model selection. The testing set was employed to evaluate the final models' performance.

F. Model Training

1. YOLOv5 Architecture: YOLOv5, a state-of-the-art object detection framework, was employed as the basis for our deep learning models. The YOLOv5 architecture offers a good balance between accuracy and speed, making it suitable for real-time monitoring applications.

2. Smoking Detection Model: The first deep learning model was trained to detect smoking instances. The training process involved feeding the annotated dataset into the YOLOv5 model, optimizing the model's parameters through backpropagation, and updating the weights accordingly. Several training iterations were performed to improve the model's accuracy.

3. Violence Detection Model: The second deep learning model was trained to detect instances of violence within the hostel environment. Like the smoking detection model, the violence detection model underwent multiple training iterations, utilizing the annotated dataset and optimizing the YOLOv5 model's parameters.

1) Evaluation

A. Performance Metrics: To assess the accuracy and effectiveness of the trained models, several performance metrics were considered. These included precision, recall, and F1-score, which provide insights into the models' ability to detect smoking and violence instances accurately [9].

B. Testing Phase: The final trained models were evaluated on the previously unseen testing set. The models' outputs were compared with the ground truth labels, and the performance metrics were calculated to measure their accuracy. Additionally, qualitative analysis of the models' detections was conducted to gain insights into their performance in real-world scenarios.

C. Comparison and Analysis: The performance results of the smoking and violence detection models were analyzed to gauge their effectiveness. The achieved accuracy rates of 93% and 94% respectively demonstrate the models' capability to accurately identify instances of interest. The limitations and potential areas for improvement were also discussed based on the evaluation results.

2) Experimental Conditions

The experimental conditions in this study are outlined in Table II. The training parameter configurations for the violence detection and smoking detection models are presented in Table III and Table IV, respectively. Both datasets, comprising the violence dataset and the smoking dataset, are divided into training, validation, and test sets in an 8:1:1 ratio.

TABLE II. EXPERIMENTAL ENVIRONMENT.

Experimental Environment	Details
Programming language	Python 3.9
Operating system	Linux Operating System
Deep-learning framework	Pytorch 1.8.2
GPU	NVIDIA Tesla K80
IDE	Google Colab
GPU Acceleration	Enabled (Google Colab with GPU)

TABLE III. TRAINING PARAMETERS FOR SMOKING DETECTION MODEL.

Training Parameters	Details
Epochs	60
Batch size	16
Image size (pixels)	640 × 640
Initial learning rate	0.01
Optimization algorithm	SGD
Pre-training weights	Yolov5n and Yolov5l Models

TABLE IV. TRAINING PARAMETERS FOR VIOLENCE DETECTION MODEL.

Training Parameters	Details
Epochs	60
Batch size	16
Image size (pixels)	640 × 640
Initial learning rate	0.01
Optimization algorithm	SGD
Pre-training weights	The best.pt obtained by training the corresponding violence detection model

TABLE V. DATASET SPLIT DETAILS.

Dataset	Train	Val	Test
Violence detection dataset	8000	2000	2000
Smoking detection dataset	4800	600	600

G. Integrating Models into a Python Flask Application

To leverage the trained deep learning models for smoking and violence detection, as well as the facial recognition component, a Python Flask application was developed [10]. The flowchart of the proposed methodology is shown in Figure 3.

This section outlines the integration process, highlighting the key components and functionalities of the application.

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

1. Application Architecture: The Python Flask framework was chosen due to its simplicity and flexibility in building web applications. The application follows a client-server architecture, where the client interacts with the web interface, and the server processes requests and performs the necessary computations using the integrated models.
2. Web Interface: The Flask application provides a user-friendly web interface that allows users to interact with the system effortlessly. The interface includes options for uploading images or videos for analysis.
3. Image and Video Processing: When a user uploads an image or video, the Flask application receives the data and passes it through the integrated deep learning models for smoking and violence detection. The models analyze the content, identify instances of smoking or violence, and generate corresponding results.
4. Result Presentation: The analysis results, including smoking and violence detections, as well as the recognized individuals, are displayed on the web interface. The Flask application generates visual representations, such as bounding boxes or labels, to highlight the identified objects or individuals in the uploaded media.

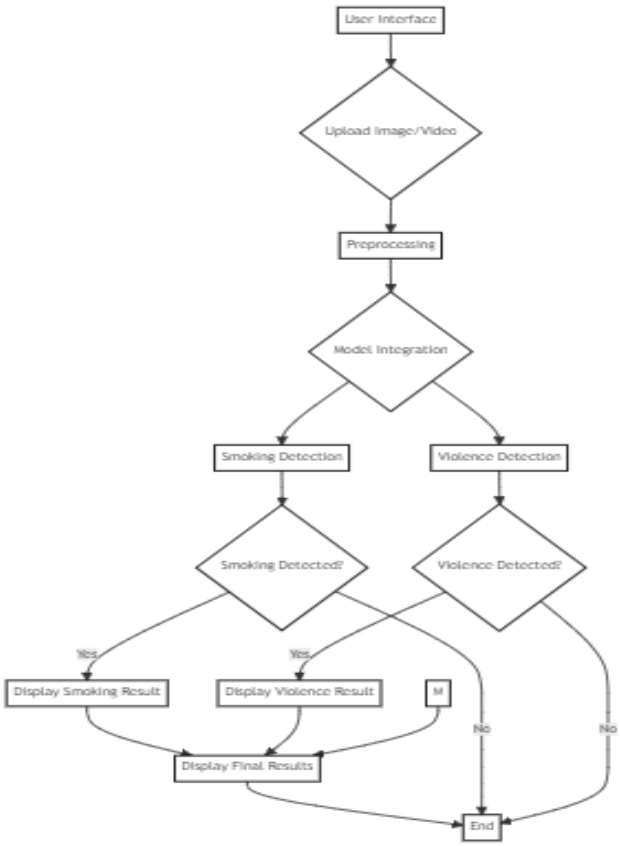

Fig. 3. Flowchart of the proposed methodology.

The Flask application provides a seamless and interactive platform for users to utilize the integrated deep learning models. By integrating the models into a user-friendly web interface, the application empowers users to analyze images and videos for smoking and violence instances, as well as perform facial recognition-based attendance tracking. The clear presentation of results enhances the usability and effectiveness of the application, contributing to improved security and monitoring capabilities in the hostel environments.

IV. RESULTS AND DISCUSSION

A. Results Analysis

This section examines the obtained data, identifies patterns, and draws meaningful conclusions, shedding light on the research objectives.

1) Overview of Model Performance

The evaluation of the integrated models within the application involves assessing their performance using various metrics [11] These metrics provide insights into the accuracy, precision, recall, and other important aspects of the models' performance. Equations 1, 2, 3, 4, and 5 show how to calculate accuracy, precision, recall, confusion matrix and f1 score, respectively.

1. Accuracy: Accuracy measures the overall correctness of the model's predictions. It is calculated by dividing the number of correct predictions by the total number of predictions made. A high accuracy indicates that the model is making correct predictions consistently.

$$Accuracy = \frac{NumberofCorrectPredictions}{TotalNumberofPredictions} x100 \qquad (1)$$

2. Precision: Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive. It focuses on the model's ability to minimize false positives. Precision is calculated by dividing the true positives by the sum of true positives and false positives.

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \times 100 \quad (2)$$

3. Recall: Recall, also known as sensitivity or true positive rate, measures the proportion of correctly predicted positive instances out of all actual positive instances. It focuses on the model's ability to minimize false negatives. Recall is calculated by dividing the true positives by the sum of true positives and false negatives.

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \times 100 \quad (3)$$

4. Confusion Matrix: The confusion matrix is a tabular representation that summarizes the performance of a classification model. It shows the number of true positives, true negatives, false positives, and false negatives. This matrix helps in visualizing the distribution of predicted and actual labels and enables the calculation of various performance metrics.

$$\begin{bmatrix} TruePositives(TP) & FalseNegatives(FN) \\ FalsePositives(FP) & TrueNegatives(TN) \end{bmatrix} \qquad (4)$$

5. F1 Score: The F1 score is a metric commonly used in classification tasks to evaluate the performance of a

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

model. It combines precision and recall into a single value that represents the balance between the two.

$$F1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (5)$$

B. Smoking Detection Results

The smoking detection model's performance is evaluated based on its ability to accurately detect instances of smoking within the application.

1. Detection Rate: The detection rate, also known as sensitivity or true positive rate, measures the proportion of correctly detected smoking instances out of all actual smoking instances.
2. False Positives: False positives refer to instances where the model incorrectly predicts smoking when there is no smoking activity present.
3. False Negatives: False negatives refer to instances where the model fails to detect smoking activity when it is present.

The smoking detection results provide insights into the model's effectiveness in identifying smoking instances and its potential for false alarms or missed detections.

In YOLOv5, the validation, train, and box loss evaluations are key metrics used to assess the performance and training progress of the object detection model. Here is a brief explanation of each evaluation:

1. Validation Evaluation: During training, the model periodically evaluates its performance on a validation dataset, which contains labeled examples that are not used for training. The validation evaluation helps to measure how well the model generalizes to unseen data. It calculates metrics such as mean average precision (Map-+), which provides an overall assessment of the model's accuracy in detecting objects across different classes.
2. Train Loss Evaluation: The train loss evaluation is used to monitor the loss function during the training process. The loss function quantifies the discrepancy between the predicted bounding boxes and the ground truth bounding boxes. It consists of various components, including classification loss, localization loss, and regularization terms. By monitoring the train loss, you can determine how well the model is learning and converging during training. The goal is to minimize the loss over time.
3. Box Loss Evaluation: The box loss evaluation specifically focuses on the localization loss component of the overall loss function. It measures the accuracy of the predicted bounding box coordinates. A lower box loss indicates that the model is effectively localizing objects in the image. This evaluation helps in assessing the model's ability to precisely determine the position and size of the objects.

These evaluations are important for monitoring the training progress, adjusting hyperparameters, and identifying potential issues or areas for improvement in the object detection model based on the observed results.

1) Evaluation Plots

During the training phase of the smoking detection model, various metrics and loss functions are monitored and visualized through training plots. These plots provide insights into the model's learning progress and can help assess its performance over time. Figures 4, 5, 6, 7 and 8 show various plots for measuring and evaluating the performance of the model after training. Here are some key aspects typically observed in smoking detection training plots:

1. F1-Confidence Curve: The F1-Confidence curve represents the relationship between the F1 score and the confidence threshold used in a classification model. It shows how the model's performance changes as the threshold for classifying positive instances is varied. The curve is useful for finding an optimal threshold that balances precision and recall in the classification task. A higher threshold may result in higher precision but lower recall, while a lower threshold may lead to higher recall but lower precision. Analyzing the F1-Confidence curve helps in determining the appropriate threshold for achieving the desired trade-off between precision and recall.

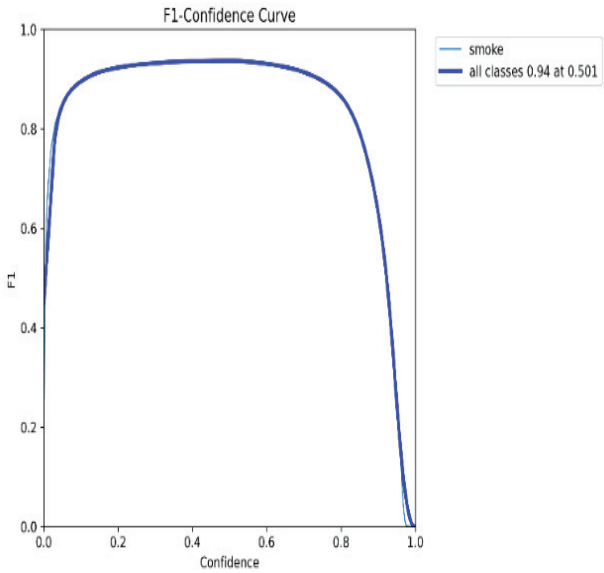

Fig. 4. F1-Confidence Curve.

2. Precision-Confidence Curve: The precision-confidence curve illustrates the relationship between precision and the confidence threshold in a classification model. It shows how precision varies as the threshold for classifying positive instances is adjusted. The curve is useful for analyzing the trade-off between precision and the number of false positives in a classification task. By adjusting the confidence threshold, one can increase precision at the cost of potentially missing some true positives or decrease precision to capture more positive instances at the risk of including false positives. The precision-confidence curve assists in identifying the optimal threshold that balances precision and the desired level of false positives in the model's predictions.

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

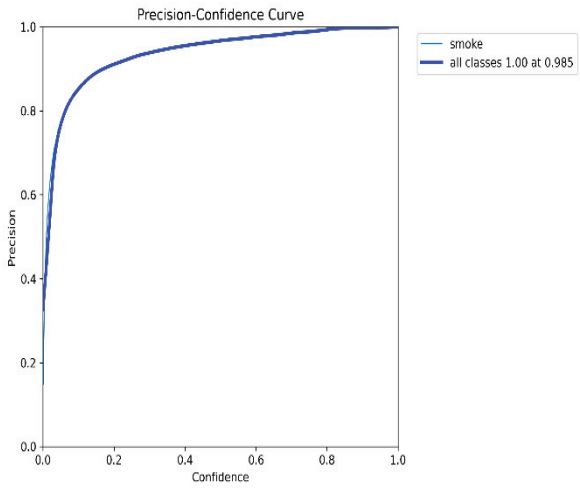

Fig. 5. Precision-Confidence Curve.

3. Precision-Recall Curve: The precision-recall curve is a graphical representation of the trade-off between precision and recall in a binary classification model. It shows how the precision and recall values change as the classification threshold is varied. The curve is useful for evaluating the model's performance across different thresholds and determining the optimal threshold for the given task.

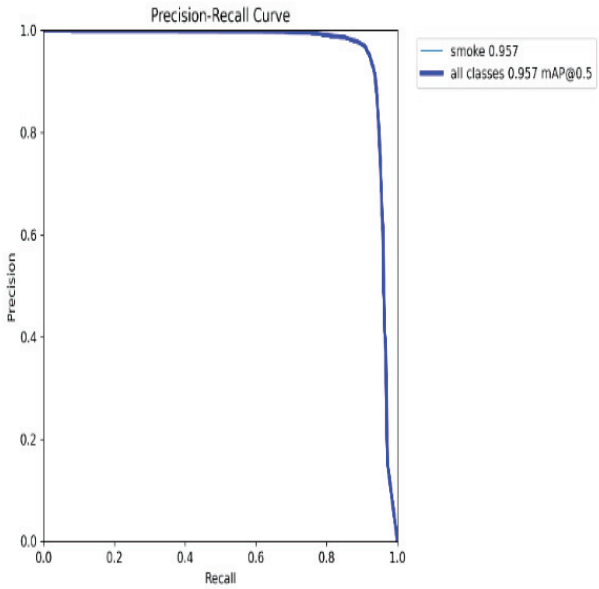

Fig. 6. Precision-Recall Curve.

4. Recall-Confidence Curve: The recall-confidence curve depicts the relationship between recall (also known as sensitivity) and the confidence threshold in a classification model. It illustrates how the recall value changes as the threshold for classifying positive instances is adjusted.

5. The curve is particularly useful for assessing the trade-off between correctly identifying positive instances (true positives) and potentially missing some positive instances (false negatives). By varying the confidence threshold, one can increase recall at the expense of potentially including more false positives. Conversely, a higher threshold may improve precision but could result in lower recall.

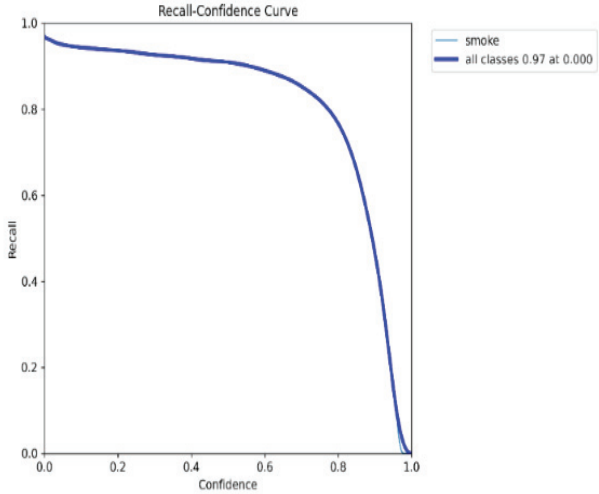

Fig. 7. Recall-Confidence Curve.

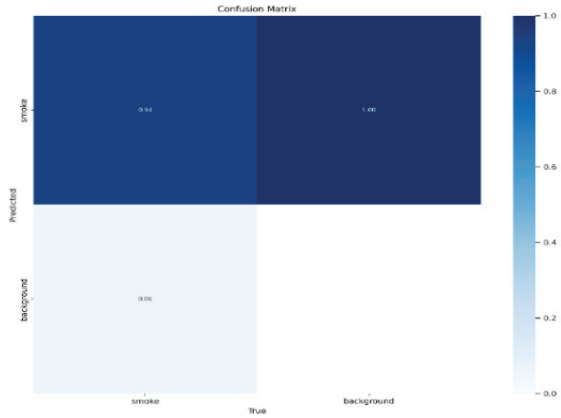

Fig. 8. Confusion Matrix.

C. *Violence Detection Results*

The violence detection model's performance is evaluated based on its ability to accurately detect instances of violence within the ABUAD HMS application. The complete results table is available in the appendix section of this report.

1. Detection Rate: The detection rate, also known as sensitivity or true positive rate, measures the proportion of correctly detected violence instances out of all actual violence instances.

2. False Positives: False positives refer to instances where the model incorrectly predicts violence when there is no violent activity present.

3. False Negatives: False negatives refer to instances where the model fails to detect violent activity when it is present.

The violence detection results provide insights into the model's effectiveness in identifying violent instances and its potential for false alarms or missed detections.

By analyzing these metrics and evaluating the smoking and violence detection results, we can assess the performance and capabilities of the integrated models within the flask application.

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.

A. Discussion

This section critically analyzes and interprets the research results, providing insights and implications for further exploration.

1) Comparative Analysis

In this section, we will compare the performance of the smoking and violence detection models within the application, shedding light on any notable differences or similarities.

The smoking detection model demonstrates a high detection rate, accurately identifying instances of smoking within the monitored environment. The false positive and false negative rates, however, require careful consideration. False positives may occur when the model wrongly detects smoking activity in non-smoking scenarios, leading to unnecessary alerts. False negatives, on the other hand, may occur when the model fails to detect smoking instances, potentially resulting in missed violations.

Similarly, the violence detection model exhibits a commendable detection rate, effectively identifying instances of violence within the monitored area. False positives and false negatives are potential areas of concern. False positives can lead to unnecessary alarms triggered by non-violent activities, while false negatives may result in undetected instances of violence.

2) Identification of Challenges

During the integration and deployment process, several challenges and limitations were encountered, which could have influenced the results obtained. These challenges include:

1. Data Quality and Quantity: The performance of the integrated models heavily relies on the quality and quantity of the training data. Insufficient or biased training data can impact the models' accuracy and generalizability.
2. Environmental Factors: Real-world scenarios present various environmental factors that can affect the models' performance. Lighting conditions, variations in facial appearances, and different smoking/violence contexts can pose challenges to accurate detection.
3. Model Optimization: Fine-tuning the models to achieve the desired balance between sensitivity and specificity can be a complex task. Adjusting thresholds and parameters requires careful consideration to minimize false positives and false negatives.

V. CONCLUSION

This paper explored the significant role of large language models (LLMs) in natural language processing (NLP) and their impact on various aspects of society. We discussed the advantages of LLMs in improving language understanding, text summarization, classification, and scalability. However, it is crucial to acknowledge the limitations, such as bias and interpretability challenges, associated with LLMs. The ethical concerns, including privacy, security, fairness, and accountability, have also been highlighted.

Future research and development should concentrate on eliminating biases in training data, enhancing the interpretability and explainability of LLMs, creating strong evaluation metrics, improving data privacy and security measures, encouraging interdisciplinary collaborations, and establishing moral frameworks for ethical LLM deployment. Researchers and practitioners may help enhance LLMs while addressing the problems they face by putting these ideas into practice.

The incorporation of LLMs into NLP and AI has extensive repercussions on societal and economic issues as well as sectors and professions associated with language. There is an obvious necessity for proactive actions to address job displacement and economic inequality while also emphasizing the potential advantages of democratizing access to knowledge. In order to successfully navigate the difficulties connected with LLMs, continuous research, ethical principles, and regulatory frameworks are vital. Ethical, legal, and justice issues are also very important.

REFERENCES

[1] S. Cheurprakobkit and R. A. Bartsch, “Security measures on school crime in Texas middle and high schools,” *Educ. Res.*, vol. 47, no. 2, pp. 235–250, 2005.

[2] T. C. Bache, S. R. Bratt, J. Wang, R. M. Fung, C. Kobryn, and J. W. Given, “The intelligent monitoring system,” *Bull. Seismol. Soc. Am.*, vol. 80, no. 6B, pp. 1833–1851, 1990.

[3] G. Jocher *et al.*, “ultralytics/yolov5: v7.0 - YOLOv5 SOTA Realtime Instance Segmentation.” Zenodo, Nov. 2022, doi: 10.5281/zenodo.7347926.

[4] A. Ortis, P. Caponnetto, R. Polosa, S. Urso, and S. Battiato, “A report on smoking detection and quitting technologies,” *Int. J. Environ. Res. Public Health*, vol. 17, no. 7, p. 2614, 2020.

[5] J. U. Ahmed, K. U. Ahmed, M. A. S. Shimul, and R. Zuñiga, “Managing strategies for higher education institutions in the UK: an overview,” *High. Educ. Futur.*, vol. 2, no. 1, pp. 32–48, 2015.

[6] D. Mali, R. T. Patil, and N. Dharwadkar, “Development of A Simple and Low Cost Unified Smart Video Surveillance System for Monitoring Hostel Student Attendance.”

[7] E. Bermejo Nievas, O. Deniz Suarez, G. Bueno García, and R. Sukthankar, “Violence detection in video using computer vision techniques,” in *Computer Analysis of Images and Patterns: 14th International Conference, CAIP 2011, Seville, Spain, August 29-31, 2011, Proceedings, Part II 14*, 2011, pp. 332–339.

[8] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “Arcface: Additive angular margin loss for deep face recognition,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4690–4699.

[9] S. Visa, B. Ramsay, A. L. Ralescu, and E. Van Der Knaap, “Confusion matrix-based feature selection.,” *Maics*, vol. 710, no. 1, pp. 120–127, 2011.

[10] P. S. Lokhande, F. Aslam, N. Hawa, J. Munir, and M. Gulamgaus, “Efficient way of web development using python and flask,” 2015.

[11] R. Padilla, S. L. Netto, and E. A. B. Da Silva, “A survey on performance metrics for object-detection algorithms,” in *2020 international conference on systems, signals and image processing (IWSSIP)*, 2020, pp. 237–242.

Authorized licensed use limited to: University of Northumbria. Downloaded on February 06,2026 at 15:49:25 UTC from IEEE Xplore. Restrictions apply.